

Support Vector Machine Analysis - Student Dropout Prediction

Understanding the Pathways to Academic Achievement

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```
## Data loaded successfully!
```

```
## Dimensions: 4424 rows x 37 columns
```

```
## First few columns: Marital.status, Application.mode, Application.order, Course, Daytime.evening.atten
```

1 Support Vector Machine Analysis

We will now test different vector models to understand what is the best classifier for our student's performance. To classify our dataset, the model plots a separating line or boundary between the two outcome classes using a number of predictors - in order to keep it clear, we want to create a two-dimensional plane (two predictors).

By applying different kernels—linear, radial, and polynomial— we assess how the shape of the decision boundary affects prediction accuracy and determine which kernel provides the most reliable classification for our dataset.

1.1 Feature Selection

For us to choose the best predictors, we will compute F-statistic ANOVA for every numeric feature in the dataset. The F-score tell us how much the mean of each feature differs between the two target groups (graduated vs. not graduated). higher scores mean there is a bigger separation between classes, and a better cutoff between groups. Results indicate that grades and the amount of credits approved have the highest F-Values, suggesting they are the most significant predictors for predicting the success of graduating or not.

In this case, we want to choose the features of second semester marks and the amount of units passed to plot our model.

Table 1: Top 5 Features by F-statistic

	Feature	F_value	p_value
31	Curricular.units.2nd.sem..approved.	1410.73	0
32	Curricular.units.2nd.sem..grade.	1134.11	0
25	Curricular.units.1st.sem..approved.	859.87	0
26	Curricular.units.1st.sem..grade.	713.52	0
17	Tuition.fees.up.to.date	505.62	0

1.2 Prepare Data for SVM

```
## Using features for SVM:

## Feature 1 (x.1): Curricular.units.2nd.sem..approved.

## Feature 2 (x.2): Curricular.units.2nd.sem..grade.

##
## Dataset prepared:

## Total rows: 4424

## Target classes: Dropout, Enrolled, Graduate
```

1.3 Exploratory Visualization

1.4 Train/Test Split

Now, we will split the data into training and testing sets, and then fit several SVM models to classify the students based on their marks.

```
## Training set size: 3541

## Test set size: 883
```

1.5 Model Training

Now, we will fit the SVM model with a linear, a radial and a polynomial kernel.

1.6 Model Evaluation

Predict and compare the models

1.6.1 Linear Kernel Results

```
##
## Linear Kernel Result:

## Confusion Matrix and Statistics
##
##           Reference
## Prediction Dropout Enrolled Graduate
## Dropout      220      65      29
## Enrolled       0       0       0
## Graduate      64      93     412
##
## Overall Statistics
##
##           Accuracy : 0.7157
##           95% CI : (0.6847, 0.7453)
##       No Information Rate : 0.4994
##       P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.4958
##
## McNemar's Test P-Value : < 2.2e-16
##
## Statistics by Class:
##
##           Class: Dropout Class: Enrolled Class: Graduate
## Sensitivity           0.7746           0.0000           0.9342
## Specificity           0.8431           1.0000           0.6448
## Pos Pred Value        0.7006           NaN           0.7241
## Neg Pred Value        0.8875           0.8211           0.9076
## Prevalence            0.3216           0.1789           0.4994
## Detection Rate        0.2492           0.0000           0.4666
## Detection Prevalence  0.3556           0.0000           0.6444
## Balanced Accuracy      0.8089           0.5000           0.7895
```

1.6.2 Radial Kernel Results

```
##
## Radial Kernel Result:

## Confusion Matrix and Statistics
##
##           Reference
## Prediction Dropout Enrolled Graduate
## Dropout      199      42      26
## Enrolled       21      23       3
## Graduate      64      93     412
```

```

##
## Overall Statistics
##
##           Accuracy : 0.718
##           95% CI : (0.6871, 0.7475)
##       No Information Rate : 0.4994
##       P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.5065
##
## McNemar's Test P-Value : < 2.2e-16
##
## Statistics by Class:
##
##           Class: Dropout Class: Enrolled Class: Graduate
## Sensitivity           0.7007           0.14557           0.9342
## Specificity           0.8865           0.96690           0.6448
## Pos Pred Value        0.7453           0.48936           0.7241
## Neg Pred Value        0.8620           0.83852           0.9076
## Prevalence            0.3216           0.17894           0.4994
## Detection Rate        0.2254           0.02605           0.4666
## Detection Prevalence  0.3024           0.05323           0.6444
## Balanced Accuracy     0.7936           0.55623           0.7895

```

1.6.3 Polynomial Kernel Results

```

##
## Polynomial Kernel Result:

## Confusion Matrix and Statistics
##
##           Reference
## Prediction Dropout Enrolled Graduate
##   Dropout      173      22      25
##   Enrolled      12       5       0
##   Graduate      99     131     416
##
## Overall Statistics
##
##           Accuracy : 0.6727
##           95% CI : (0.6407, 0.7036)
##       No Information Rate : 0.4994
##       P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.406
##
## McNemar's Test P-Value : < 2.2e-16
##
## Statistics by Class:
##
##           Class: Dropout Class: Enrolled Class: Graduate
## Sensitivity           0.6092           0.031646           0.9433
## Specificity           0.9215           0.983448           0.4796

```

## Pos Pred Value	0.7864	0.294118	0.6440
## Neg Pred Value	0.8326	0.823326	0.8945
## Prevalence	0.3216	0.178935	0.4994
## Detection Rate	0.1959	0.005663	0.4711
## Detection Prevalence	0.2492	0.019253	0.7316
## Balanced Accuracy	0.7653	0.507547	0.7115

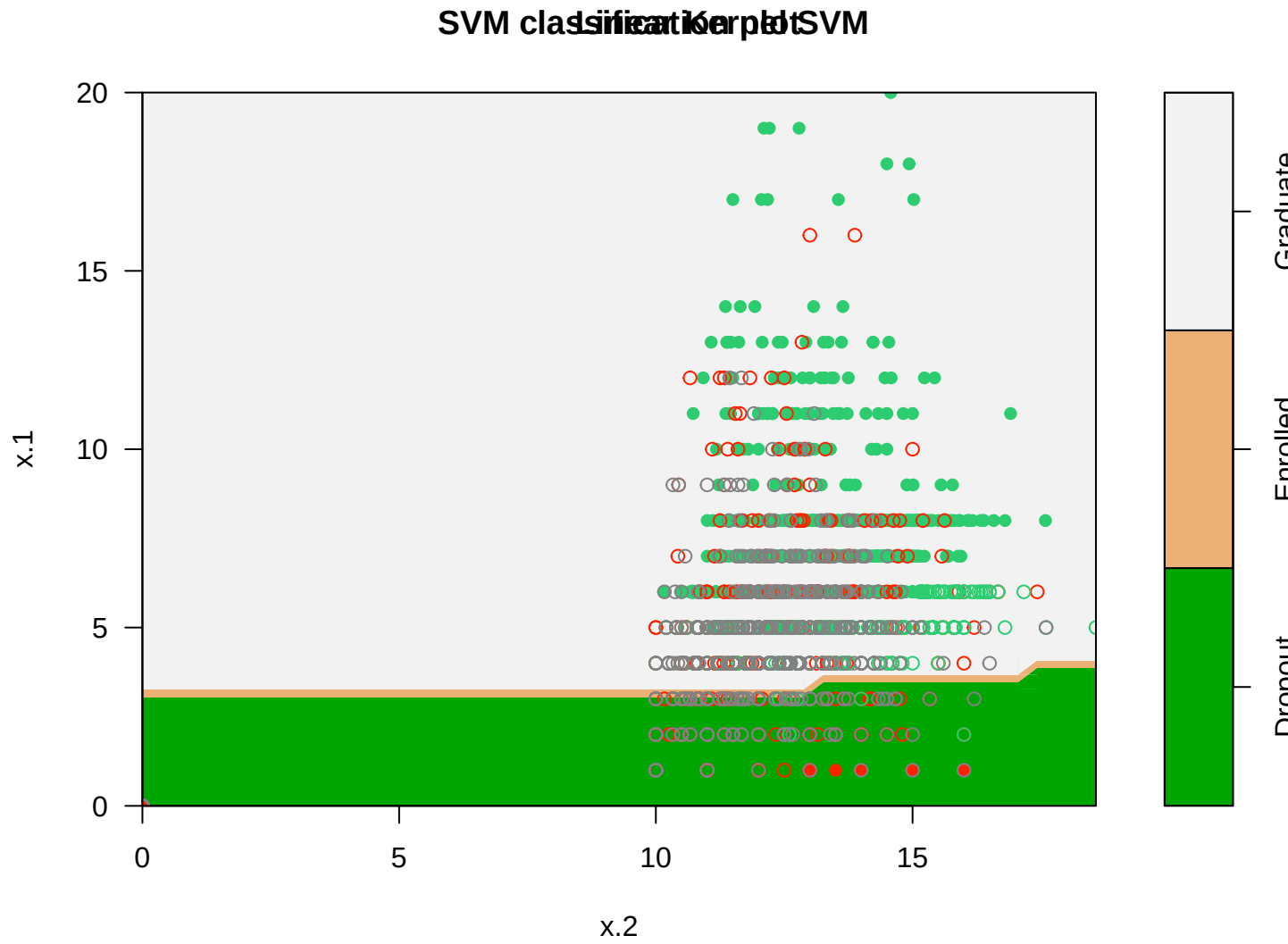
1.7 Performance Comparison

Table 2: Model Accuracy Comparison

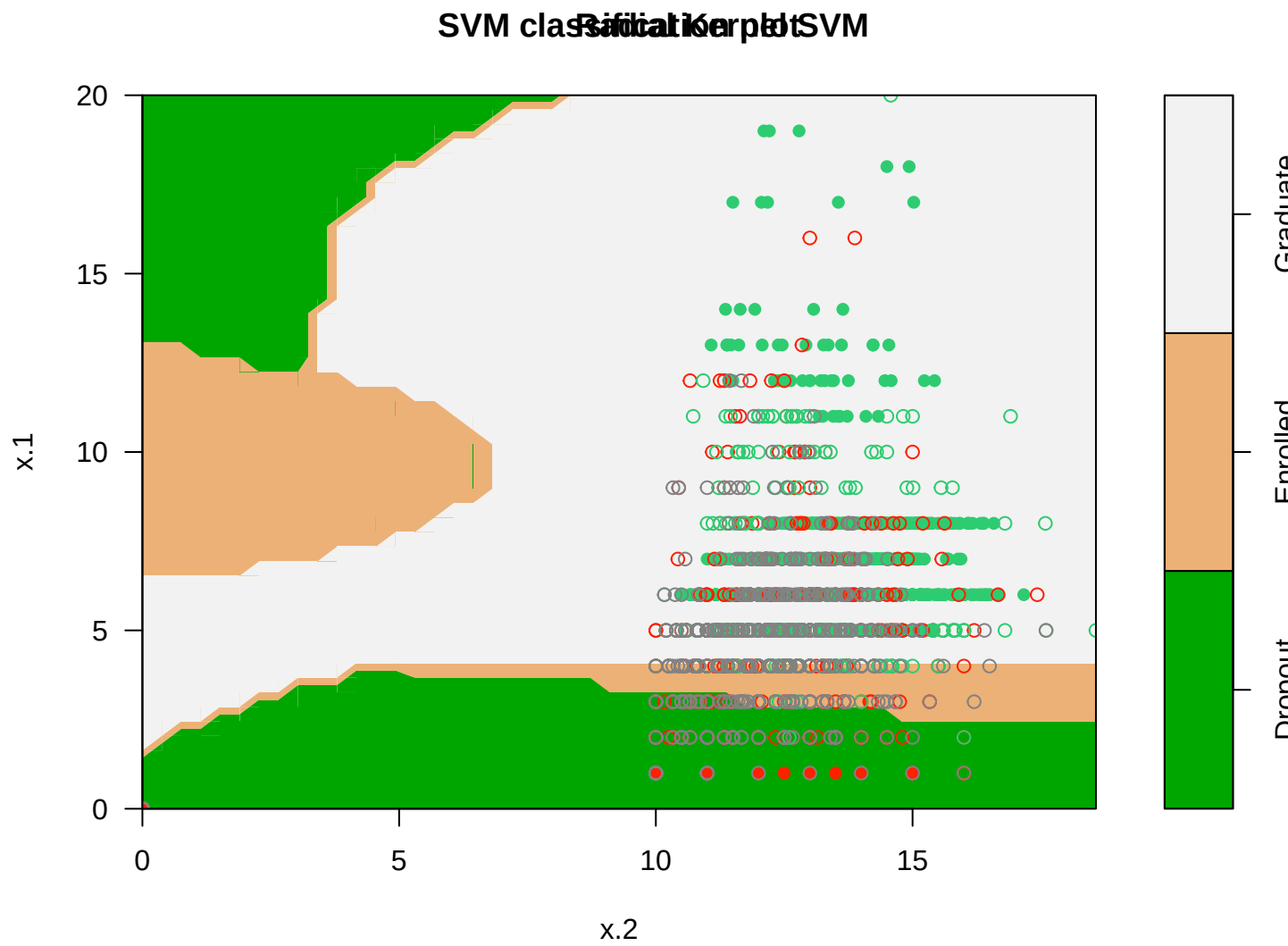
Model	Accuracy
Linear	0.716
Radial	0.718
Polynomial	0.673

1.8 Decision Boundary Visualizations

1.8.1 Linear Kernel



1.8.2 Radial Kernel



1.8.3 Polynomial Kernel

